

TEJAS AI

The AI factory testing environment

तेजस् • On-prem • Sovereign • Built for GPU halls

Train autonomous control • Stress the twin • Repair in VR • Deploy when ready

AI factories broke the old playbook

Seconds to act • Weeks of jobs at risk

Rack power spikes before inlet temp — you have seconds, not minutes

One cooling trip = ₹2+ Cr lost compute + SLA penalties

Phaidra proved autonomous control on NVIDIA-scale halls

Everyone else still runs 1990s rule-based BMS logic

30–40%
of power = cooling

The real hall is not a lab

- ✗ Live: failed job = customer churn → ✓ Twin: heat wave in one drag
- ✗ Live: wrong setpoint = chip throttle → ✓ Twin: chiller trip — recover in sim
- ✗ Live: no rollback for the board → ✓ Shadow: log would-do vs your BMS
- ✗ Live: MoC / insurance nightmare → ✓ Trust ladder with instant revert

"Would you let an untested AI drive your GPUs on Thursday's 47°C day?"

Tejas = your AI factory testing environment

Full physics twin · Zero risk to silicon

Calibrated digital twin — topology + historian, not a drawing toy

Autonomous brain vs baseline — live PUE 1.3 vs 1.5 at 47°C

Stress: heat waves · load spikes · chiller trips · GPU faults

Graduate path: Twin → Shadow (read-only) → Advise → You take control

DEMO NOW

1. /datacenter — Autonomous Control • PUE live
2. Weather → 47°C — baseline panics • Tejas holds
3. "Is any machine going down?" → GPU-16 red
4. Hand phone → /vr — stand inside the hall

What the testing environment trains

Feed-forward control — the AI factory requirement

Acts on rack power + weather before inlet spikes (<10s)

Explains every move — heat wave COP protection

Dual run: Tejas vs dumb BMS — savings gap visible live

Shadow on real hall: their historian proves ₹ before control

You graduate — not us

From testing env to real hall on your schedule

**Twin
(test)**

**Shadow
(read-only)**

**Advise
(you apply)**

**Supervised
(confirm)**

**Autonomous
(you own)**

We hand you the keys when YOUR data says yes.

Repair in VR

Less downtime · Less cost · Never walk blind

2:14 AM alarm

→ GPU-16 flagged 14 months early

Senior tech on a plane

→ VR walk to exact rack — junior on site

Open rack blind

→ Exploded design — filter, fan tray, busbar

₹2+ Cr outage

→ 25-min planned fix · parts pre-staged

VR repair flow

Walk in • Pick • Inspect • Fix • Blue

Scan QR → stand inside live twin (/vr)

Walk to GPU-16 — pulsing red • reticle pick

Aspect lenses — cooling path: filter + fan tray

Field AR on site — 4-step checklist • rack turns blue

Desktop syncs — work order closed automatically

Predict in twin • Practice in VR • Fix on site

Inlet 3.4°C hotter than load explains — 14-month creep

Root cause: CRAC-2 filter + rear fan-tray bearing

Work order + email with exact parts drafted automatically

Caught in testing twin → validated on historian in shadow

What the testing env proves

12 MW GPU hall • before one setpoint hits silicon

₹1.8–3.2 Cr

cooling saved/yr

₹2+ Cr

one avoided outage

₹8.3L+

per PM fix

₹90–160 Cr

50-hall fleet

Built for halls Phaidra doesn't serve — yet

GPU colo / edge AI DC — sovereign on-prem testing sandbox

Enterprise AI lab — internal team owns control policy

DC adding GPU block — test liquid + air mix before go-live

ESCO / integrator — white-label testing env for customers

Pay for proof — not promises

Twin build / testing env — paid on-ramp (credited to Y1)

Shadow 30 days — read-only · would-do report · low / free

Advisory + control — savings-share 20–40% when THEY graduate

Land-and-expand: one hall → campus → 50-site fleet

Don't learn autonomous cooling on live GPUs.

Learn it in the twin.

Repair it in VR.

Own the real hall when you're ready.

30 days read-only on your historian.

We build your testing environment — and show you the rack you'd lose on a 47°C Thursday.